



MACHINE LEARNING FOR MATERIAL DESIGN- MS43040

Physics-consistent stress-strain modeling of anisotropic hyperelastic heart tissue

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1 Introduction

The human heart is one of the most mechanically complex organs in the body; it undergoes large-amplitude, rhythmic deformations over billions of cycles. Accurate modeling of the constitutive behavior of the myocardium is vital for diagnosing pathologies, planning surgical interventions, and developing regenerative therapies.

The myocardium exhibits a highly anisotropic, nonlinear mechanical response that arises directly from its mesoscale structure. Cardiac muscle cells (cardiomyocytes) are organized into locally parallel fibre bundles that rotate across the ventricular wall, creating a three dimensional architecture with distinct material directions. The tissue is significantly stiffer along the fibre axis than transverse to it. This directional dependence is nonlinear; transitioning from a compliant regime at low strains to a stiff, fibre-dominated regime at large deformations. In addition, the myocardium is nearly incompressible due to its high fluid content. Together, these characteristics mean any constitutive model must simultaneously respect anisotropy, nonlinearity, and incompressibility; a combination that makes both experimental characterization and mathematical modelling challenging.

Historically, researchers have relied on phenomenological models to describe heart tissue. Well-known frameworks, such as the Holzapfel-Ogden (2009) model [1], use hand-crafted exponential equations to capture the tissue’s passive response. While these models are mathematically elegant and physically grounded, they are often limited by a fixed functional form that may not capture the full diversity of experimental data across different species or disease states.

With the recent surge in computational power, machine learning has emerged as a potential alternative. Traditional ”black-box” neural networks can process vast amounts of data to find hidden patterns; however, they face two critical limitations in biomechanics:

1. **Data Scarcity:** High-quality mechanical data from human cardiac tissue is notoriously difficult to obtain, and standard neural networks require massive datasets to learn basic physics from scratch.
2. **Physical Unreliability:** A standard neural network optimizes purely for data fit, with no awareness of the underlying physics. Without explicit constraints, the network may converge to solutions that violate fundamental physical laws.

To bridge this gap, we implement a Constitutive Artificial Neural Network (CANN). By embedding the fundamental laws of continuum mechanics directly into the network architecture, we move beyond simple curve-fitting. The CANN performs Automated Model Discovery, selecting the most accurate terms from a library of physically admissible functions, combining the flexibility of machine learning with the rigor of continuum mechanics.

2 Problem Description

Soft biological tissues are hyperelastic: they sustain large, finite deformations and recover elastically upon unloading. Unlike linear elastic solids, hyperelastic materials exhibit a nonlinear stress-strain relationship, where stiffness increases with strain. In their passive state, such materials are fully characterized by a scalar strain energy function Ψ , from which all stresses are derived.

The myocardium is additionally orthotropic, meaning its mechanical properties differ along three mutually perpendicular anatomical axes:

1. **Fibre direction** (f): Aligned with the long axis of the myocytes; the stiffest direction.
2. **Sheet direction** (s): Parallel to the myocardial laminae, allowing for interlaminar sliding.
3. **Normal direction** (n): Perpendicular to the sheet planes, representing the wall thickness direction.

Furthermore, the myocardium is nearly incompressible due to its high fluid content, meaning volumetric changes under physiological loading are negligible. Any constitutive model must therefore satisfy hyperelasticity, orthotropy, and incompressibility, alongside broader thermodynamic requirements such as objectivity and polyconvexity. The following sections establish the mathematical foundation of continuum mechanics and detail how each of these constraints is embedded into the CANN.

2.1 Continuum Mechanics Foundation

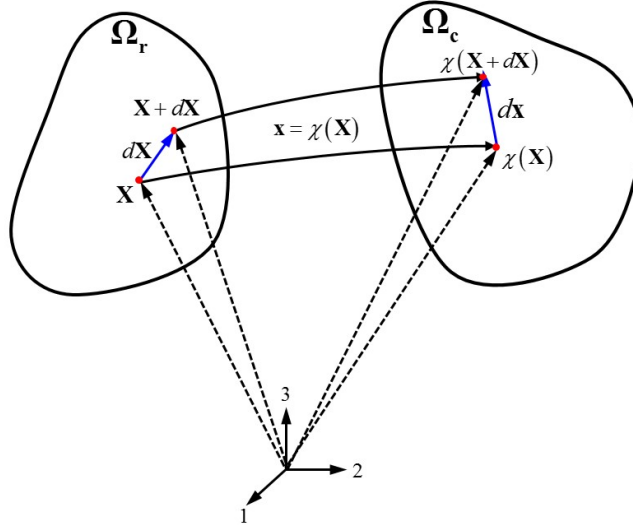


Figure 1: Mapping of a line segment under deformation (adapted from [2]).

For a given line segment $d\mathbf{X}$ in the reference configuration (Ω_r), its deformed state in the current configuration (Ω_c) is given by $d\mathbf{x}$ and can be described by the deformation gradient $\mathbf{F}$ such that

$$d\mathbf{x} = \mathbf{F} d\mathbf{X}, \quad \mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}. \quad (2.1)$$

The deformation gradient contains complete information about local dimensional changes and expresses both rigid-body rotation and stretches. Its determinant, $J = \det(\mathbf{F})$, represents a local volume change and is typically constrained for incompressible materials to $J = 1$.

As rigid-body rotation included in $\mathbf{F}$ will not contribute to a resulting strain energy, a common objective form of the strain tensor is given by the Right Cauchy-Green tensor,

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} \quad (2.2)$$

In using this form which is unchanged throughout a rigid-body rotation, a convenient basis is established for invariant-based strain energy density representations of material deformation. For finite strains in hyperelastic materials, deformations can be expressed as a function of the stored elastic strain energy density $\Psi(\mathbf{F})$. The stress response represented by the 1st Piola-Kirchhoff stress tensor P_{ij} follows directly from differentiation of this strain energy by deformation gradient F_{ij} :

$$P_{ij} = \frac{\partial \Psi}{\partial F_{ij}} \quad (2.3)$$

More commonly, Ψ is modeled as a function of rotational invariants of $\mathbf{C}$ such that,

$$\Psi(\mathbf{F}) \rightarrow \Psi(I_1, I_2, I_3) \quad (2.4)$$

where,

$$\begin{aligned} I_1 &= \text{tr}(\mathbf{C}), \\ I_2 &= \frac{1}{2} [(\text{tr}(\mathbf{C}))^2 - \text{tr}(\mathbf{C}^2)], \\ I_3 &= \det(\mathbf{C}) \end{aligned} \quad (2.5)$$

Consequently, stress is obtained as:

$$P_{ij}(\mathbf{F}) = \frac{\partial \Psi}{\partial I_1} \frac{\partial I_1}{\partial F_{ij}} + \frac{\partial \Psi}{\partial I_2} \frac{\partial I_2}{\partial F_{ij}} + \frac{\partial \Psi}{\partial I_3} \frac{\partial I_3}{\partial F_{ij}} \quad (2.6)$$

When considering materials with directionally-dependent properties, several invariant conditions change and further complexity is introduced. A more detailed explanation of stress invariants as they relate to anisotropic material systems is presented in section 4.2.

3 Modeling Goals

The primary modeling goal is automated constitutive model discovery: to identify, from the experimental data alone, which physically admissible strain-energy terms are necessary and sufficient to reproduce the observed stress–stretch behaviour across all 11 loading protocols simultaneously.

Specifically, we aim to:

1. Discover a sparse set of active invariant–function pairs that explain the data without overfitting.
2. Ensure thermodynamic consistency: non-negative energy, zero energy at the reference state, and polyconvexity.
3. Quantify the trade-off between model complexity (number of active terms) and prediction accuracy via a Pareto analysis.
4. Assess generalisation using cross-validation, to confirm that the discovered model captures genuine physics rather than artifacts of the training set.

4 Approach: CANN Architecture

To model the passive behavior of heart tissue, a CANN is implemented; a schematic overview is shown in Figure 2. The network takes the deformation gradient $\mathbf{F}$ as input and returns the first Piola-Kirchhoff stress $\mathbf{P}(\mathbf{F})$ as output. The forward pass proceeds in three stages.

First, $\mathbf{F}$ is mapped to the eight physically motivated invariants K_i described in Section 4.2, which form the actual inputs to the network. Each invariant is then passed through two candidate activation functions: a polynomial term $(o)^n$ for $n \in \{1, 2\}$ and an exponential term $\exp(o)$. Together they produce a library of 32 candidate terms in total. Each candidate term is scaled by a learnable weight $w_{1,i}$, followed by a second set of weights $w_{2,i}$ before being summed into the strain energy density $\Psi(\mathbf{F})$. Finally, the stress is obtained by automatic differentiation of Ψ with respect to $\mathbf{F}$:

$$\mathbf{P}(\mathbf{F}) = \frac{\partial \Psi}{\partial \mathbf{F}}. \quad (4.1)$$

This structure means the CANN does not learn a free-form mapping, but instead performs automated model selection within a fixed library of physically meaningful expressions. The physical constraints embedded in this architecture are detailed in the following subsections.

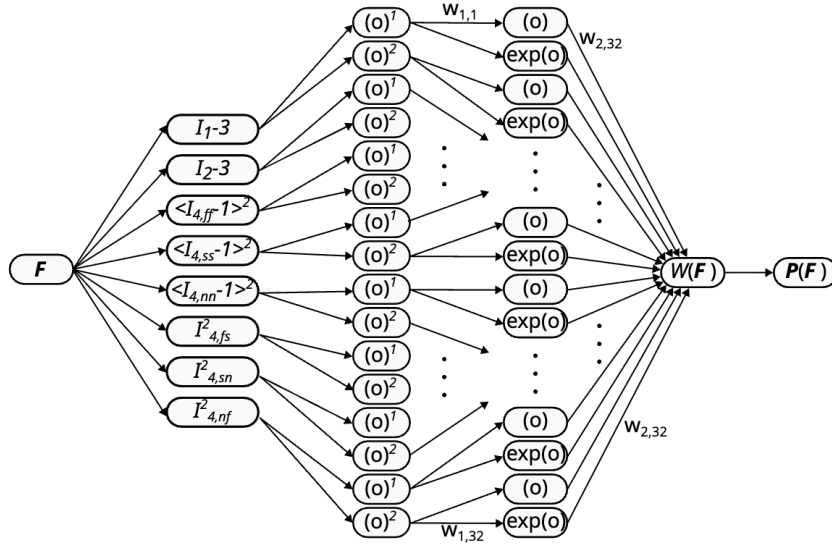


Figure 2: Schematic overview of the Constitutive Artificial Neural Network architecture for discovering the behavior of heart tissue [3].

4.1 Material Objectivity

A fundamental requirement of any constitutive model is that the predicted stress must be independent of the observer's frame of reference. In other words, a rigid-body rotation of the entire body should produce no change in the stored strain energy. This property is known as material objectivity, or frame indifference. The raw deformation gradient $\mathbf{F}$ contains rigid-body rotations that carry no physical energy. By using the Right Cauchy-Green tensor $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ as the network input instead, frame indifference is guaranteed by construction.

4.2 Orthotropy

Material symmetry ensures that the model's predictions are consistent with this internal architecture. In a standard black box neural network, the model might accidentally learn that the heart is stiffer when

pulled North than when pulled South. By enforcing symmetry through invariants, these directional biases are avoided. The three anatomical axes (f , s , n) of the myocardium give rise to additional anisotropic invariants. The components of $\mathbf{C}$ are mapped to eight invariants K_i that capture the isotropic matrix response (K_1, K_2), axial fibre stretching (K_4), and shear coupling (K_8):

- Isotropic: $K_1 = I_1 - 3$, $K_2 = I_2 - 3$
- Axial stretch: $K_{4f} = \langle I_{4,ff} - 1 \rangle^2$, $K_{4s} = \langle I_{4,ss} - 1 \rangle^2$, $K_{4n} = \langle I_{4,nn} - 1 \rangle^2$
- Shear coupling: $K_{8fs} = I_{8,fs}^2$, $K_{8sn} = I_{8,sn}^2$, $K_{8nf} = I_{8,nf}^2$

The invariants are shifted by their reference values (e.g. $I_1 = 3$ in the undeformed state), ensuring $\Psi = 0$ when there is no deformation, consistent with conservation of energy. The Macaulay brackets $\langle \cdot \rangle$ ensure fibres contribute to the strain energy only under tension. When a fibre is compressed ($I_{4,ff} < 1$), the bracket returns zero and the term vanishes from Ψ , capturing fibre buckling under compression.

4.3 Incompressibility

Since the myocardium is nearly incompressible, the constraint $J = \det(\mathbf{F}) = 1$ is imposed, $I_3 = J^2 = 1$ carries no information and is excluded from Ψ . For incompressible materials, the hydrostatic pressure is not constitutively determined and must be treated separately via a Lagrange multiplier, giving the augmented stress:

$$\Psi_{\text{total}} = \Psi(\mathbf{C}) - p(J - 1) \quad (4.2)$$

The indeterminate hydrostatic pressure p is determined by exploiting the experimental boundary condition $P_{nn} = 0$; the tissue is unloaded in the sheet-normal direction in the biaxial tests, which uniquely fixes p and corrects all remaining stress components accordingly.

4.4 Polyconvexity and Thermodynamic Stability

To guarantee that Ψ is thermodynamically stable, two measures are taken:

1. Non-negative weights: A softplus activation ensures all network weights $w \geq 0$, preventing energy from decreasing with increasing deformation.
2. Convex activation functions: Only linear, quadratic, and exponential functions are used as energy kernels. Since these are summed with non-negative weights and no coupling terms between invariants, Ψ is a non-negative linear combination of convex functions and therefore convex by construction.

5 Methodology

5.1 Data Description

The model is trained and validated using a dataset of human myocardial tissue mechanics acquired by Sommer et al. (2015) [4]. To capture the full orthotropic nature of the heart, the dataset consists of 11 independent mechanical protocols across two test types:

1. **Biaxial Extension (5 protocols):** Tension is applied simultaneously along the fiber (f) and normal (n) directions at varying stretch ratios (e.g., 1:1, 0.5:1, 1:0.75). The directly measured quantities are the stretches F_{ff} , F_{nn} and the corresponding stresses P_{ff} , P_{nn} . Since the tissue is not loaded in the sheet direction, the remaining stress components are assumed to be zero. The sheet stretch F_{ss} is not measured but is recovered from the incompressibility constraint, since the tissue cannot change volume:

$$F_{ss} = \frac{1}{F_{ff} F_{nn}}. \quad (5.1)$$

2. **Triaxial Shear (6 protocols):** Simple shear deformations are applied across all six anatomical planes (fs , fn , sf , sn , nf , ns). For a shear test on plane i with loading in direction j , the measured quantities are the shear component $F_{ij} = \gamma$ and the corresponding stress component P_{ij} .

The biaxial and shear measurements are assembled into the full 3×3 deformation gradient tensor $\mathbf{F}$, which serves as the input to the CANN. Because $J = \det(\mathbf{F}) = 1$ is imposed, the invariant $I_3 = J^2 = 1$ carries no information and is excluded from Ψ .

The experimental dataset exhibits the characteristic features expected of passive cardiac tissue (see Figures 3 and 4).

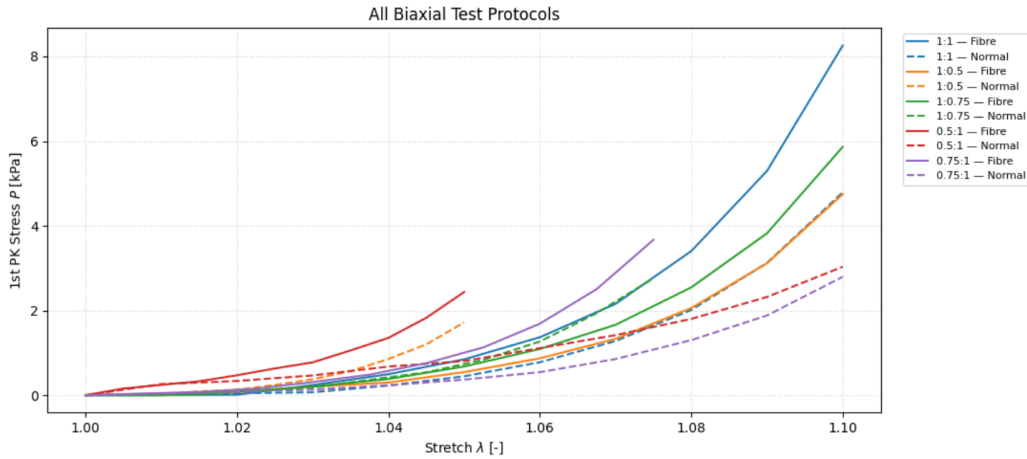


Figure 3: Raw biaxial test data for all five stretch-ratio protocols. Solid lines: fibre direction (P_{ff}); dashed lines: normal direction (P_{nn}). The fibre direction is markedly stiffer and becomes more so as the ratio shifts toward normal-dominant loading, demonstrating strong mechanical coupling.

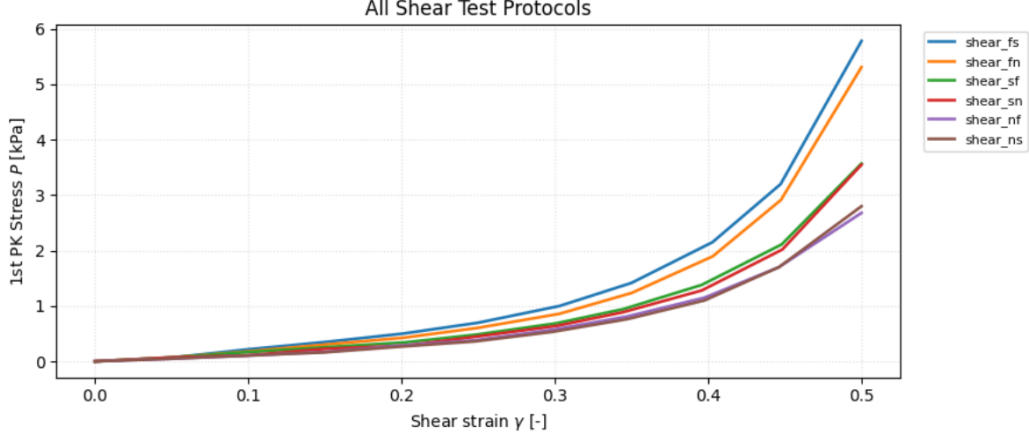


Figure 4: Raw triaxial shear data for all six anatomical planes. The pronounced asymmetry between modes (e.g. *sf* and *ns* are markedly stiffer than *fs* and *nf*) motivates the use of separate anisotropic shear invariants K_{8ij} in the constitutive library.

Under equibiaxial loading (1:1), the fibre direction is markedly stiffer than the normal direction. As the ratio shifts toward normal-dominant loading (e.g. 1:0.5), fibre-direction stiffness increases further, demonstrating the strong mechanical coupling inherent to the tissue’s microstructure. The six shear modes display pronounced asymmetry: resistance to sliding is highly direction-dependent, with *sf* and *ns* showing substantially higher stiffness than *fs* and *nf*, motivating the use of separate anisotropic shear invariants.

5.2 Training Procedure

The network is trained by minimizing the mean squared error between the predicted and experimentally measured stress components across all 11 protocols simultaneously. Only the physically measured stress components contribute to the loss: the fibre and normal components (P_{ff} , P_{nn}) for biaxial tests, and the single shear component (P_{ij}) for each shear protocol. Unobserved components are excluded so that the model is not penalised for predicting physically consistent but unmeasured values. The loss function is defined as:

$$\mathcal{L} = \underbrace{\text{MSE}(P_{ff}^{\text{pred}}, P_{ff}^{\text{exp}}) + \text{MSE}(P_{nn}^{\text{pred}}, P_{nn}^{\text{exp}}) + \sum_{\text{shear modes}} \text{MSE}(P_{ij}^{\text{pred}}, P_{ij}^{\text{exp}})}_{\mathcal{L}_{\text{MSE}}} + \alpha_{L1} \sum |w| + \alpha_{L2} \sum w^2 \quad (5.2)$$

where α_{L1} and α_{L2} are regularization coefficients controlling the sparsity of the learned weights. In the initial training run, both are set to zero, meaning all 32 candidate terms are free to contribute to Ψ .

The role of regularization in promoting sparsity is discussed in the following section. Optimization is performed using the L-BFGS algorithm with a learning rate of 0.1, a history size of 10, and the strong Wolfe line search condition. L-BFGS is a quasi-Newton (second-order curvature) method that approximates the inverse Hessian from the last 10 gradient vectors. For this problem, a small dataset, a smooth twice-differentiable loss, and only 64 parameters, this yields convergence in roughly an order of magnitude fewer function evaluations than first-order methods such as Adam, making it well-suited to the limited experimental data available here. The strong Wolfe conditions guarantee both sufficient decrease and a curvature condition at each step, which is critical for the tight loss-surface valleys associated with exponential energy terms having large curvature parameters. Because L-BFGS may evaluate the closure multiple times per outer step during line search, the loss is re-evaluated inside a closure at each iteration. An early-stopping criterion (relative improvement threshold $\tau = 10^{-7}$,

patience of 8 steps) halts training once the loss has converged, substantially reducing unnecessary computation. The model typically converges within 8 epochs, as shown in Figure 5.

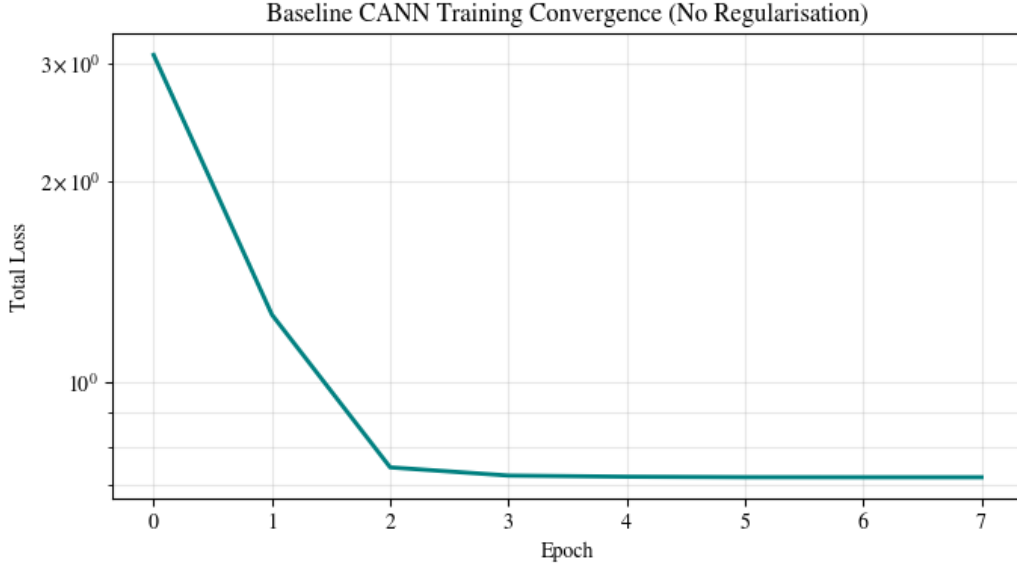


Figure 5: Training loss as a function of epoch, plotted on a logarithmic scale. The L-BFGS optimiser converges rapidly (within 8 epochs) validating the use of early stopping.

5.3 Regularization and Sparsity

To promote both interpretability and generalizability, two forms of regularization are incorporated into the training objective 3. The loss function 5.2, contains two penalty coefficients α_{L1} and α_{L2} that balance the minimization of prediction error against the regularization objectives.

L1 Regularization (Lasso) promotes sparsity by penalizing the absolute sum of the network weights. This encourages the optimizer to drive weak, physically irrelevant terms to zero, leaving only the most significant contributions to the strain energy density. Formally:

$$\mathcal{L}_{L1} = \alpha_1 \sum_j |w_j| \quad (5.3)$$

However, due to the strictly positive nature of the softplus activation and the large steps taken by the L-BFGS optimizer, L1 alone cannot drive weights to exactly zero in practice. A hard thresholding step is therefore applied after training, permanently setting any weight below a threshold $\tau = 0.005$ to zero:

$$w_j \leftarrow \begin{cases} w_j & \text{if } w_j > \tau \\ 0 & \text{if } w_j \leq \tau \end{cases} \quad (5.4)$$

The threshold $\tau = 0.005$ is physically justified: at this level a term contributes less than 0.05 % of the maximum observed stress (≈ 10 kPa), making it negligible for all practical purposes.

L2 Regularization (Ridge) promotes numerical stability by penalizing the squared sum of the network weights. This prevents any single term from dominating the strain energy through an excessively large weight, which is particularly important for exponential activation functions where large weights can cause numerical overflow. Formally:

$$\mathcal{L}_{L2} = \alpha_2 \sum_j w_j^2 \quad (5.5)$$

The optimal α_{L1} and α_{L2} are selected via a Pareto analysis, described in the following section.

5.4 Model Selection via Pareto Analysis

A key challenge in regularized model discovery is selecting the optimal regularization strength α_1 : too small and the model retains redundant terms; too large and physically meaningful terms are suppressed. To identify the best compromise between sparsity and accuracy, we scan over a logarithmically spaced grid of values,

$$\alpha_1 \in \{0, 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 10^{-1}\}, \quad (5.6)$$

combined with a secondary grid for the L2 stability coefficient:

$$\alpha_2 \in \{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}\}. \quad (5.7)$$

For each (α_1, α_2) pair, a fresh CANN is trained from scratch, hard thresholding is applied, and two quantities are recorded: (i) the number of active terms n_w after thresholding, and (ii) the mean squared error on the training data. A term is considered active only when the product $w_1 \cdot w_2$ exceeds the threshold of 0.005 after hard thresholding: a large w_1 with a negligible w_2 would produce a near-zero energy regardless of the invariant, and vice versa, so the product criterion correctly identifies dormant terms. The Pareto table records only the data-fidelity component $\mathcal{L}_{\text{MSE}}$, not the full regularised loss. Recording the regularised loss would conflate the penalty with the prediction error, making high- α models appear artificially worse. By always evaluating the same pure-MSE metric, every point on the Pareto front is comparable on the same scale regardless of the α used during training.

Table 1 reports the MSE and active term count for a representative subset of (α_1, α_2) pairs. Across the range $\alpha_2 \in \{0, 10^{-8}, 10^{-6}, 10^{-4}\}$ the MSE values and active term counts are nearly identical for every fixed α_1 : for instance, at $\alpha_1 = 10^{-3}$ the MSE varies by less than 0.1 % across these four values of α_2 , and the active term count remains stable. The first appreciable degradation appears at $\alpha_2 = 10^{-3}$, where the MSE begins to rise and active term counts drop noticeably across all α_1 levels. We therefore fix $\alpha_2 = 10^{-4}$: the largest value that leaves both accuracy and sparsity unaffected, providing a small degree of numerical stabilisation without distorting the solution.

Table 1: Diagnostic sweep: MSE and number of active terms for selected (α_1, α_2) pairs. Values highlighted in grey mark the chosen operating point.

α_2	α_1	MSE	Active terms	Note
0	10^{-3}	7.205×10^{-1}	6	
0	5×10^{-3}	7.557×10^{-1}	4	
10^{-8}	10^{-3}	7.204×10^{-1}	6	
10^{-8}	5×10^{-3}	7.564×10^{-1}	4	
10^{-6}	10^{-3}	7.203×10^{-1}	7	
10^{-6}	5×10^{-3}	7.568×10^{-1}	4	
10^{-4}	10^{-3}	7.208×10^{-1}	7	chosen α_2
10^{-4}	5×10^{-3}	7.640×10^{-1}	4	chosen operating point
10^{-3}	10^{-3}	7.359×10^{-1}	5	degradation onset
10^{-3}	5×10^{-3}	7.830×10^{-1}	4	
10^{-2}	10^{-3}	1.048×10^0	3	over-regularised
10^{-2}	5×10^{-3}	1.049×10^0	3	

With $\alpha_2 = 10^{-4}$ fixed, the optimal α_1 is identified from the Pareto front in Figure 6. As α_1 increases from 0 to 10^{-1} , the model complexity falls from 13 active terms to 2, while the MSE rises from 7.203×10^{-1} to 1.502. The curve exhibits a clear elbow at 4 active terms ($\alpha_1 = 5 \times 10^{-3}$): beyond this

point further sparsification yields a disproportionate increase in error, whereas below it the additional terms produce only marginal accuracy gains.

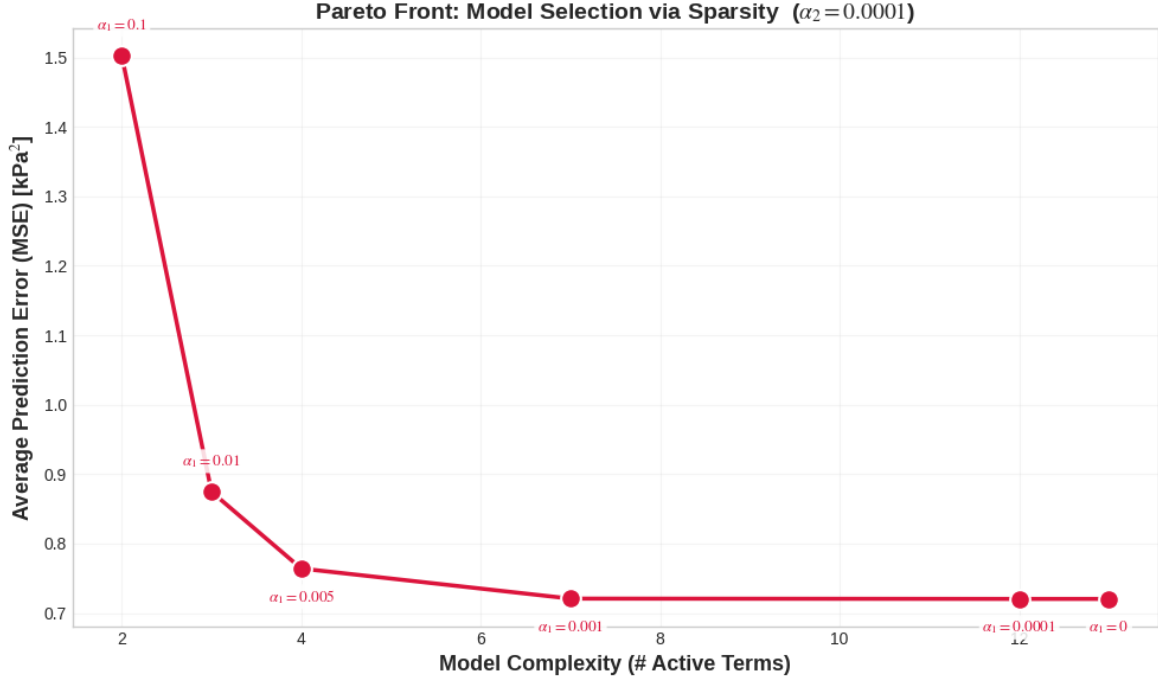


Figure 6: Pareto front for fixed $\alpha_2 = 10^{-4}$: model complexity (number of active terms after hard thresholding) versus pure prediction error (MSE without regularisation penalty). Each point is labelled with its corresponding α_1 value. The elbow at 4 active terms ($\alpha_1 = 5 \times 10^{-3}$) identifies the optimal sparse model: further sparsification causes a disproportionate accuracy loss.

We therefore select $\alpha_1 = 5 \times 10^{-3}$, which reduces model complexity by 9 terms relative to the dense baseline ($\alpha_1 = 0$, 13 active terms) while incurring an MSE increase of only approximately 5%. This marginal accuracy cost confirms that the discarded terms were fitting noise rather than genuine physical behaviour.

6 Results

6.1 Baseline Model (No Regularisation)

With $\alpha_1 = \alpha_2 = 0$, the CANN uses all 32 candidate terms and achieves a high overall fit to both biaxial and shear data. Figure 7 shows the complete constitutive fingerprint: the stacked-area decomposition reveals which energy terms carry the mechanical response across all 17 protocols. The dominant contributions come from the isotropic K_1 , K_2 terms (blue shades) and the anisotropic fibre term K_{4f} (cyan/green), consistent with the known microstructural hierarchy of cardiac tissue. Across all protocols, the R^2 values reported in each panel of Figure 7 are uniformly close to unity, confirming that the 32-term library is sufficiently expressive to reproduce both the biaxial coupling and the shear asymmetry without systematic bias. Many of the 32 terms remain nearly invisible across all protocols, providing direct visual motivation for sparsification: these terms are contributing nothing beyond noise. The transition from this overfit baseline to the sparse model discovered via the Pareto analysis is therefore both statistically and physically justified.

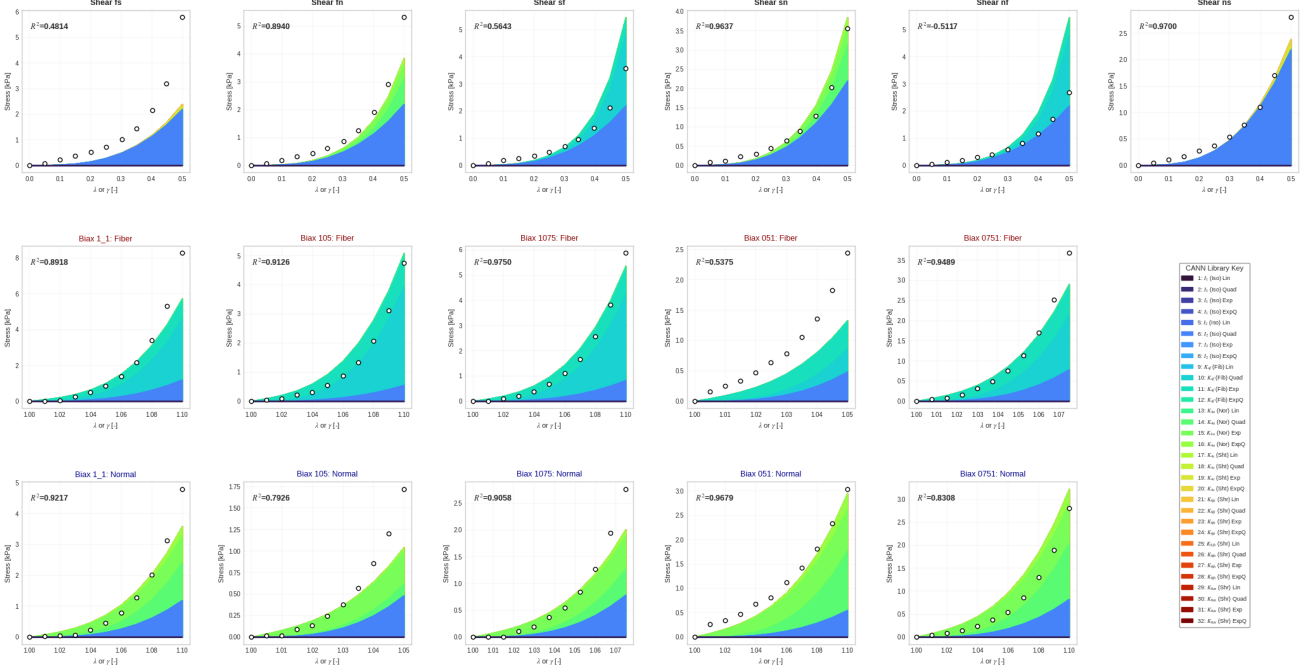


Figure 7: Baseline constitutive fingerprint ($\alpha_1 = \alpha_2 = 0$): stacked-area decomposition of the 32 energy-library terms across all 17 loading protocols (6 shear, top row; 5 biaxial fibre, middle row; 5 biaxial normal, bottom row). White circles are experimental observations; the R^2 value for each protocol is printed in the upper-left corner. Colours encode individual library terms as shown in the legend.

6.2 Discovered Sparse Model

The Pareto analysis (Figure 6) identifies the elbow at **4 active terms** ($\alpha_1 = 5 \times 10^{-3}$, $\alpha_2 = 10^{-4}$). The discovered model retains:

#	Invariant	Function	w_1	w_2
1	K_2	Quadratic	2.1467	2.1330
2	K_{4f}	Quadratic	6.3609	6.3778
3	K_{4f}	Exponential	0.9809	1.1690
4	K_{4n}	Exponential	0.4637	3.7418

The physical interpretation is clear: the matrix response is captured by the quadratic K_2 term (isotropic

stiffness); fibre recruitment and stiffening are captured by both a quadratic and an exponential K_{4f} term; and the coupling between the fibre and normal directions is described by the exponential K_{4n} term.

The co-selection of both a quadratic and an exponential term for K_{4f} is physically meaningful, not redundant. At low strains, collagen fibres are crimped and progressively recruited into load-bearing alignment, a process well described by a quadratic (polynomial) response. At higher strains, once fibres are fully taut, the tissue enters an exponential strain-stiffening regime driven by the intrinsic elasticity of the straightened collagen. The simultaneous presence of both activation functions therefore allows the CANN to capture the complete, two-phase mechanical behaviour of the fibre architecture — from the compliant toe-region of collagen uncrimping to the stiff fibre-dominated regime — without any prior specification of this functional form.

The discovered strain energy function is therefore:

$$\Psi = w_{1,1}w_{2,1}K_2^2 + w_{1,2}w_{2,2}K_{4f}^2 + w_{1,3}(e^{w_{2,3}K_{4f}} - 1) + w_{1,4}(e^{w_{2,4}K_{4n}} - 1). \quad (6.1)$$

This structure is strikingly similar to the Holzapfel–Ogden (2009) model [1], which also uses exponential fibre terms as its primary building blocks. The fact that the CANN recovers an analogous form purely from data, without any prior specification of functional form, simultaneously validates the physics-based model and demonstrates that the CANN framework can recover physically meaningful constitutive laws automatically.

Figure 8 shows the constitutive fingerprint of the sparse model. Despite using only 4 terms, the model captures the essential features of all protocols with R^2 values comparable to the 32-term baseline across the majority of protocols, confirming the physical adequacy of the discovered equation. Where modest reductions in R^2 are observed, most notably in certain shear modes, these correspond to protocols whose physical behaviour is dominated by the shear invariants K_{8ij} , which are intentionally eliminated by the Lasso penalty. The fact that these accuracy penalties are small and localised confirms that the discarded shear terms were capturing noise rather than fundamental constitutive physics.

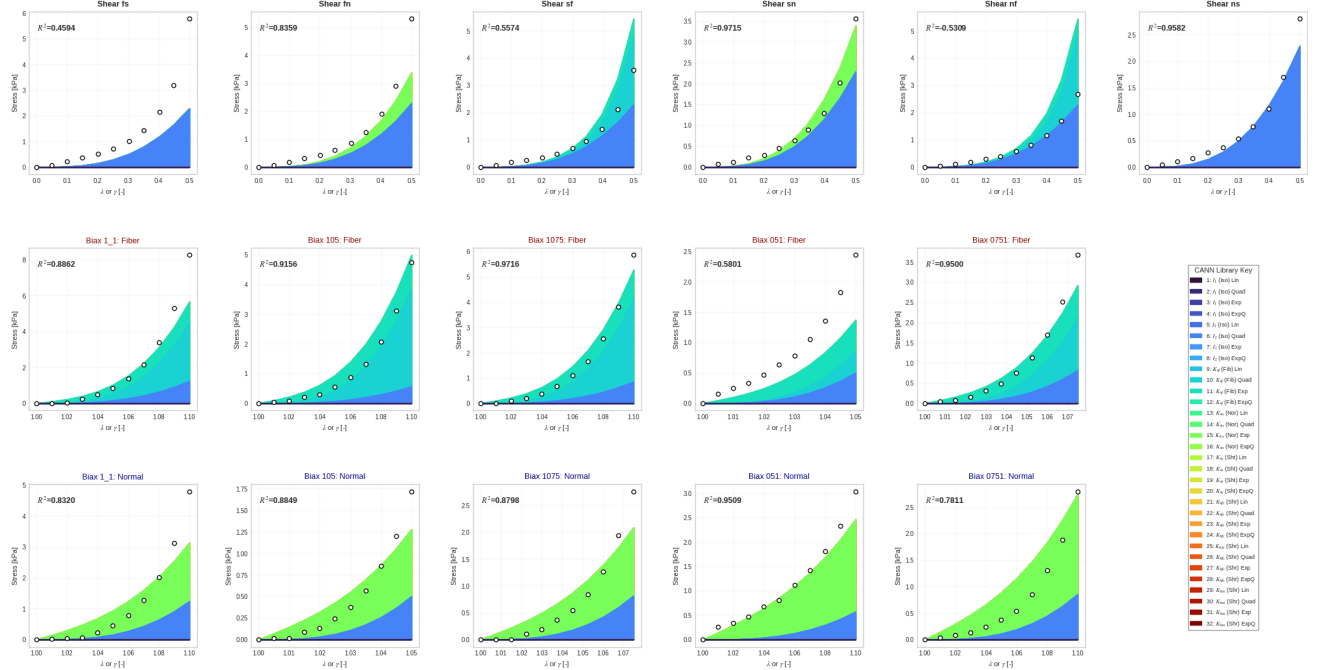


Figure 8: Sparse model constitutive fingerprint ($\alpha_1 = 5 \times 10^{-3}$, $\alpha_2 = 10^{-4}$, 4 active terms). Layout as in Figure 7. The reduction from 32 to 4 terms introduces only a modest increase in prediction error while substantially improving interpretability.

6.3 Effective Weight Distribution

Figure 9 contrasts the effective weight magnitudes ($w_1 \cdot w_2$) of the baseline and sparse models. In the baseline, multiple terms carry appreciable weight, indicating redundancy. After sparsification, only four terms survive, with K_{4f} -Quad dominating (≈ 12), confirming that fibre-direction stiffness is the principal mechanical feature of the tissue.

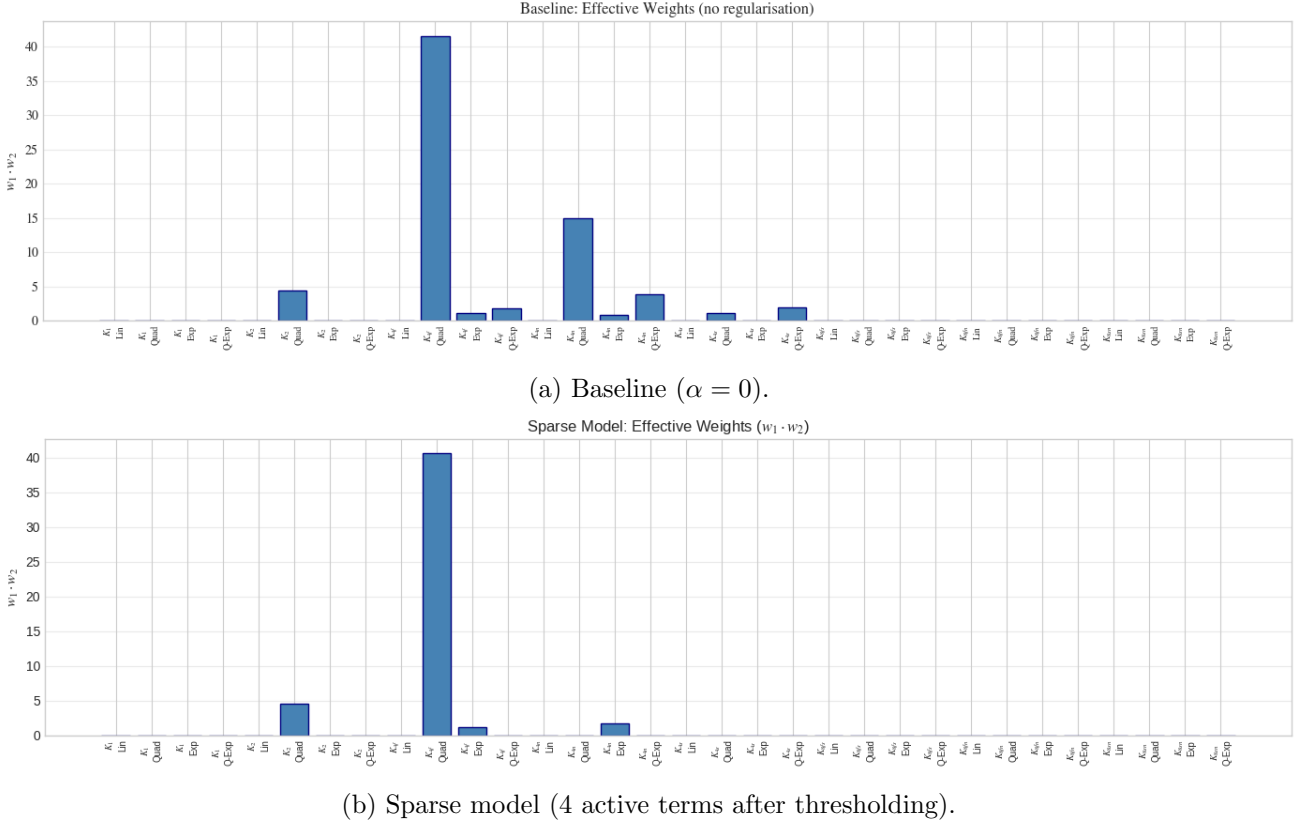


Figure 9: Effective weight magnitudes $w_1 \cdot w_2$ per library term. The K_{4f} -Quad term dominates in both models, confirming fibre-direction stiffness as the primary mechanical feature.

6.4 Refinement

Once the optimal sparse model was identified via the Pareto analysis, a refinement stage was explored. The weights of the inactive terms were frozen at zero and the regularisation penalty was removed, allowing the optimiser to retrain exclusively on the remaining active parameters without the restrictive pull of the Lasso penalty.

Although this procedure was implemented and tested, it was ultimately not adopted as the final evaluation strategy. The reason is a form of optimistic bias, closely analogous to data leakage in supervised learning: the sparsity structure, i.e. which of the 32 candidate terms survive, was itself determined by fitting the full training set. Re-optimising the weights of those same surviving terms on the same data means the model has effectively seen the data twice: once to perform model selection (which terms to keep), and once to fit those terms without any penalisation. Any in-sample accuracy metric (R^2 , MSE) reported after refinement therefore overestimates generalisation performance, because it implicitly credits the model for choices that were made by memorising the training data. This prevents fair comparison across regularisation levels on a common evaluation footing. For this reason, all final model evaluations use the regularised training loss and the un-refined sparse model weights, which were never re-exposed to the data after the sparsity structure was fixed.

7 Validation

The sparsest model selected through Pareto analysis in section 4 was determined to be the optimal discovered model which balances loss accuracy and interpretability. Despite this, it cannot be determined how well the model generalizes from training loss alone. Due to the sparsity of the dataset and the constitutive architecture of the model, even small perturbations of the training set can have critical impact on which groupings of terms survive regularization. In this way, cross-validation (CV) methods such as k-fold CV prove to be useful less as illustrations of generalize-ability, but rather as methods to probe the sensitivity of the input space. In an effort to analyze this space, two complementary CV methods were employed: stratified Monte-Carlo (MC-CV)—which perturbs the data on the scale of individual data points—and leave-one-out (LOO-CV)—which perturbs the data on the scale of entire loading condition protocols. The use of both methods seeks to answer two questions: whether the model is sensitive to specific regimes within a testing protocol, and whether the model is stable under the removal of entire loading modes.

7.1 Stratified Monte Carlo Cross-Validation

Stratified MC is a method in which each of the 11 input protocols is independently subjected to a split of 80-20% (training and validation points) and repeated across 10 random folds. In this way, it is guaranteed that sampling occurs across all input protocols, and thus isolates the effect of which stress-strain points are withheld from training. The model discovered in Section 4 is retrained during each random fold using the same L-BFGS optimization and regularization strengths. The MSE on the held out validation data is recorded as the error for a given fold.

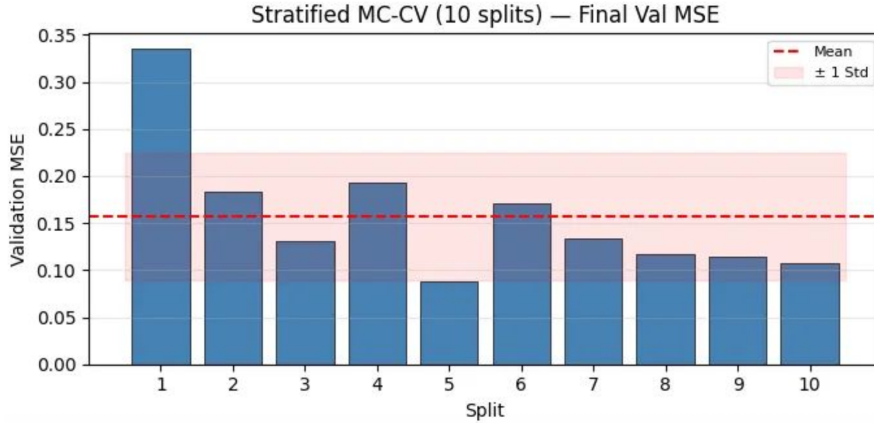


Figure 10: Final validation MSE across 10 stratified Monte Carlo folds. The red dashed line marks the mean across folds and the shaded band indicates ± 1 standard deviation.

Figure 10 shows that the validation MSE across folds varies by approximately a factor of four from ~ 0.09 to ~ 0.34 for the best and worst folds, respectively. Because there is equal representation of loading protocols across the folds, the observed variability must be attributed to *which points within a protocol* are being withheld. Further inspection of the outlier folds confirms this: the highest validation errors tend to occur when the largest-stretch points in the stiffest biaxial protocols are withheld from the training set. Physically, this means that the model is most affected when it loses the training signal for the stiff, fiber-dominated response regime. When this occurs, we observe instability in the coupled terms as the model tries to compensate for this behavior.

7.2 Leave-One-Out Cross-Validation

To observe how input sensitivity is affected at the scale of loading protocols, leave-one-out (LOO) CV is performed in which entire data protocols are withheld from the training set. The model is refit on the remaining ten sets and validation error is reported when trying to predict for the held-out data. Rather than reporting a single validation error as in the above exercise, this LOO implementation tracks how the network weights in each layer adjust from fold to fold. A term which has values clustered across all 11 folds is well-constrained by the dataset regardless of withheld training protocol. In contrast, a term whose values vary greatly across folds is likely physically described only by certain protocols or combinations thereof.

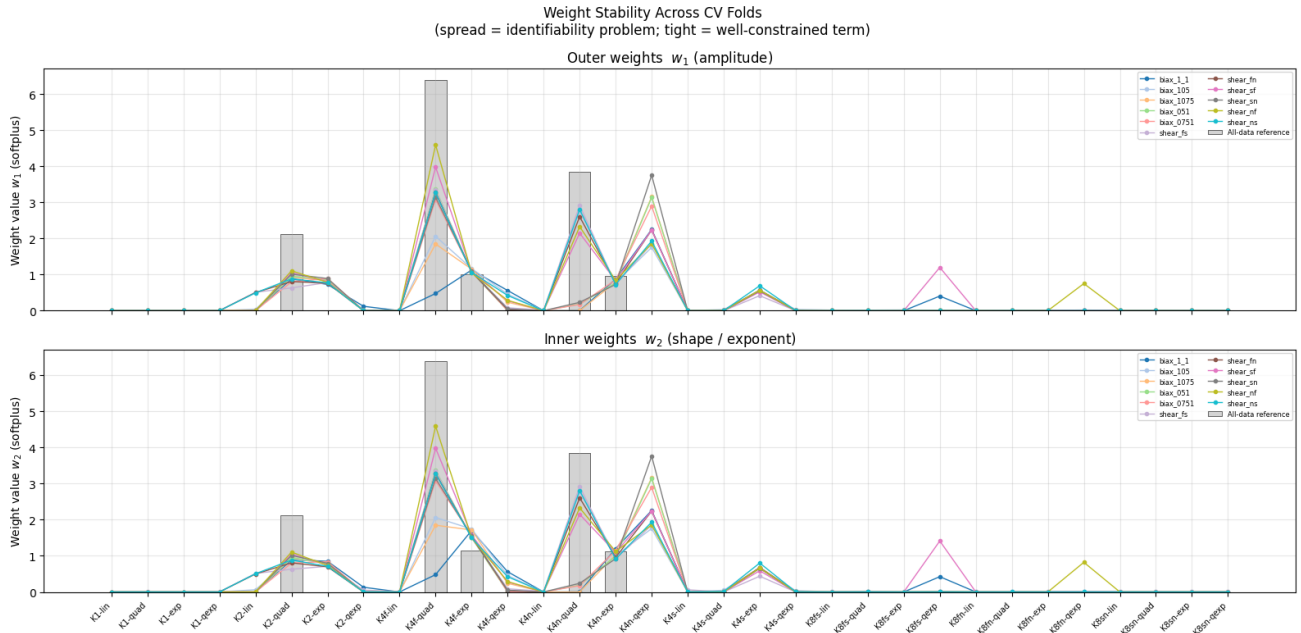


Figure 11: Stability of the outer weights w_1 (top) and inner weights w_2 (bottom) across 11 LOO folds. A grey bar overlays the graphs which represents the weights determined by the sparse model. Each line corresponds to a held out protocol; tight clustering indicates a well-constrained term, while a lone spike indicates a term whose magnitude is controlled almost exclusively by a single protocol.

Figure 11 shows that the dominant surviving terms from regularization— $K_{4f,\text{quad}}$ and $K_{4f,\text{exp-}}$ —are tightly clustered across random folds and lends confirmation that the Pareto-determined fiber and shear-coupling terms are reliable features of the heart tissue. The fold values (lines) are consistently lower than the all-data reference (bars) for these features. This same pattern also arises in features $K_{4n,\text{quad}}$ and $K_{4n,\text{exp}}$, and perhaps suggests there is some synergy between features that only arises across the entire data space. Fiber term stability produces consistent outliers through the biax 1:0.5 hold-out. This loading protocol imposes the most asymmetric fiber-dominated stretch and as such heavily constrains the value of the K_{4f} exponential contribution.

Shear-coupling invariants K_{8fs} , K_{8fn} , and K_{8sn} are small or inactive in the all-data reference, but can exhibit spikes in weight importance throughout certain hold-out folds. This behavior indicates a compensation occurring: when a given shear or biaxial loading mode is removed, the optimizer attempts to activate shear-coupling terms in order to make up for the lost constraint. The patterns are closely mirrored in w_1 and w_2 panels and suggests that the amplitude and curvature of each term are jointly stable or jointly unstable.

8 Physical Interpretation

Based on the dominant terms determined through regularization and the subsequent CV analyses, a consistent through-line is established which shows a robustness of structural content in the discovered model. The fiber-direction (K_{4f}) and normal-direction (K_{4n}) show strong recovery across every fold of both CV analyses with non-zero weights. The quantitative stability—how *strongly* each term contributes—is notably less robust. At the individual data point level, MCCV uncovers that holding out high-stretch tail points of the stiffer biaxial protocols will disproportionately inflate validation error. At the loading protocol scale, LOO reveals that sheet direction term K_{4s} and isotropic term K_2 contributions are the least stable. Secondary shear-coupling terms also tend to activate irregularly in order to compensate during individual folds.

Conclusive physical interpretation using these methods seems to reveal a limitation of the dataset rather than the architecture of the CANN itself. It follows from a strongly non-linear response represented with a handful of measurements that some protocols and regimes disproportionately weight the loss. From this it can be said that the discovered model is well-constrained in its *form* but should be calibrated over a larger data space. If the provided model were implemented into a finite-element framework, LOO weight variability could be propagated to the final output as to determine some uncertainty of the results.

9 The Big Picture

9.1 Constitutive Modeling of Cardiac Tissue: State of the Art

The passive mechanical behaviour of the myocardium has been studied extensively. Fung-type exponential models were among the first to capture the nonlinear stiffening response of soft tissues. The landmark Holzapfel–Ogden (2009) model [1] provided a rigorous thermodynamically consistent framework tailored to the orthotropic architecture of the heart, using exponential functions of the structural invariants $I_{4,ff}$, $I_{4,ss}$, and $I_{8,fs}$. Sommer et al. (2015) [4] provided the comprehensive experimental dataset used in this work, characterising the full three-dimensional passive response of human ventricular myocardium through the biaxial and shear protocols described above.

The present CANN approach recovers a constitutive law that is structurally similar to the Holzapfel–Ogden model - with exponential fibre terms playing a central role — but discovers this form from data rather than from a priori physical assumptions. This demonstrates that physically principled machine learning can both validate established phenomenological models and potentially identify novel terms that those models miss.

9.2 Clinical and Engineering Applications

Accurate constitutive models of myocardial tissue are a key enabling technology for *in silico* clinical trials. Patient-specific finite element models of the heart, informed by medical imaging and calibrated with data-driven constitutive laws, allow clinicians to:

- Predict post-operative remodelling and optimise surgical planning for conditions such as hypertrophic cardiomyopathy.
- Design and virtually test cardiac assist devices, including left ventricular assist devices (LVADs) and transcatheter aortic valve replacements (TAVRs).
- Personalise ablation therapy planning for arrhythmia treatment by accurately simulating mechanical stress distributions.

- Accelerate the regulatory approval of novel cardiac devices by reducing the number of physical prototype tests required.

The CANN framework is particularly well-suited to this clinical context because its interpretable, sparse output directly yields a symbolic constitutive equation that can be implemented in any finite element code, without requiring the trained neural network to be embedded in the simulation.

9.3 Limitations and Future Work

The current study has several limitations that motivate future work:

1. **Limited dataset:** The model is calibrated on a single dataset from one donor. Generalisation across donors, disease states, and species requires training on larger, pooled datasets.
2. **Passive response only:** The CANN models the purely passive mechanical response. Active contraction, viscoelasticity, and damage mechanics are not yet accounted for.
3. **Identifiability:** The cross-validation results in Notebook 2 indicate that some minor weight terms carry a high coefficient of variation across random splits, suggesting that a small number of invariant terms are not robustly constrained by the current experimental data. Additional loading protocols or higher data density in the toe-region may be required to uniquely identify all terms.
4. **Threshold sensitivity:** The hard-thresholding value $\tau = 0.005$ was selected on physical grounds. A more principled approach - for instance Bayesian evidence or information-theoretic model selection - could replace this heuristic in future work.

10 Conclusion

This work demonstrated that a Constitutive Artificial Neural Network can automatically discover a physically meaningful constitutive law for passive human myocardial tissue from a limited experimental dataset. By embedding the fundamental constraints of continuum mechanics; objectivity, orthotropy, incompressibility, and thermodynamic stability directly into the network architecture, the CANN combines the flexibility of machine learning with the rigour of physics-based modelling.

Through a sparsity-promoting L1 regularisation sweep and Pareto analysis, the model reduced a library of 32 candidate terms to just 4 active contributions: a quadratic isotropic term (K_2), a quadratic and an exponential fibre term (K_{4f}), and an exponential normal-direction term (K_{4n}). This sparse model achieves comparable accuracy to the overparameterised baseline while remaining fully interpretable. Notably, the discovered structure closely resembles the Holzapfel-Ogden (2009) model, validating both the CANN framework and the established phenomenological literature without any prior specification of functional form.

Cross-validation confirmed that the model is robust to perturbations at the level of individual data points, though sensitivity increases when the high-stretch fibre-dominated regime is underrepresented in training. Leave-one-out validation further revealed that most active terms are well-constrained across loading protocols, with only minor instability in the coupled invariants.

The results highlight both the promise and the current limitations of data-driven constitutive modelling. The present work models only the passive, rate-independent response of tissue from a single donor. Future work should extend the framework to active contraction and to viscoelasticity, which captures how the tissue responds differently at different loading rates. Combined with larger, multi-donor datasets, these extensions would position the CANN as a practical tool for patient-specific cardiac simulations and computer-based clinical trials.

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